

VECTOR CALCULUS AND VECTOR INTEGRATION

1. If $\phi = x^2 + y - z - 1$, then $|\nabla \phi|$ at $(1, 0, 0)$ is

- (a) $\sqrt{2}$ (b) $\sqrt{6}$ (c) $\sqrt{5}$ (d) 1

2. The directional derivative of $\phi = 3x^2 + 2y - 3z$ at $(1, 1, 1)$ in the direction of $2\hat{i} + 2\hat{j} - \hat{k}$ is

- (a) $\frac{13}{3}$ (b) 4 (c) $\frac{19}{3}$ (d) none of these

3. The unit normal vector to the surface $x^2 + y^2 - 2z + 3 = 0$ at $(1, 2, -1)$

- (a) $\frac{2\hat{i}+4\hat{j}+2\hat{k}}{\sqrt{24}}$ (b) $\frac{2\hat{i}+4\hat{j}}{\sqrt{20}}$ (c) $\frac{2\hat{i}+2\hat{j}}{\sqrt{20}}$ (d) $\frac{2\hat{i}+4\hat{j}-2\hat{k}}{\sqrt{24}}$

4. If $f = \tan^{-1}\left(\frac{y}{x}\right)$ then $\text{div}(\text{grad } f)$ is equal to

- (a) 1 (b) -1 (c) 0 (d) 2

5. The value of $\text{curl}(\text{grad } f)$, where $f = 2x^2 - 3y^2 + 4z^2$ is

- (a) $4x - 6y + 8z$ (b) $4x\hat{i} - 6y\hat{j} + 8z\hat{k}$ (c) 0 (d) 3

6. What is the value of $\nabla(xy\hat{i} + yz\hat{j} + zx\hat{k})$ is

- (a) $-y\hat{i} + z\hat{j} - x\hat{k}$ (b) $-y\hat{i} - z\hat{j} - x\hat{k}$
(c) $-y - z - x$ (d) $-y + z - x$

7. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then $\text{div}\vec{r} = \underline{\hspace{2cm}}$ and $\text{curl}\vec{r} = \underline{\hspace{2cm}}$

- (a) $\hat{i} + \hat{j} + \hat{k}$ and 0 (b) 3 and $\vec{0}$ (c) 3 and 0 (d) none of these

8. The angle between the vectors $6\hat{i} + 2\hat{j} + 3\hat{k}$ and $2\hat{i} - 9\hat{j} + 6\hat{k}$ is

- (a) $\theta = 1.4143$ (b) $\theta = 0.897$ (c) $\theta = 0$ (d) $\theta = 0.1558$

9. If $\vec{F} = xyz\hat{i} + 3x^2y\hat{j} + (xz^2 - y^2z)\hat{k}$, then at $(2, -1, 1)$ $\nabla \cdot \vec{F} = \underline{\hspace{2cm}}$

- (a) $-\hat{i} + 12\hat{j} + 3\hat{k}$ (b) $-\hat{i} + 12\hat{j} + 5\hat{k}$ (c) 16 (d) 14

10. Find 'a' such that $(-x^2 + yz)\hat{i} + (4ay - z^2x)\hat{j} + (2xz - 4z)\hat{k}$ is solenoidal

- (a) -1 (b) 1 (c) 0 (d) none of these

11. Find 'a' such that the vector $\vec{F} = (x + y + az)\hat{i} + (x + 2y - z)\hat{j} + (-x - y + 2z)\hat{k}$ is irrotational

- (a) -1 (b) 1 (c) 0 (d) none of these

12. If $\phi = xy^3z^2 = 4$, then $\nabla \phi$ at the point $(1, 1, -1)$ is

- (a) $\hat{i} + 3\hat{j} + 2\hat{k}$ (b) $\hat{i} + 3\hat{j} - 2\hat{k}$
(c) $\hat{i} - 3\hat{j} + 2\hat{k}$ (d) $\hat{i} - 3\hat{j} - 2\hat{k}$

13. The conservative vector field is

- (a) irrotational (b) solenoidal
(c) both solenoidal and irrotational (d) neither solenoidal nor irrotational

14. The unit directional derivative to the curve $x = t, y = t^2, z = t^3$ at the point $(-1, 1, -1)$ is

- (a) $\frac{1}{\sqrt{14}}(\hat{i} - 2\hat{j} + 3\hat{k})$ (b) $\frac{1}{\sqrt{14}}(\hat{i} + 2\hat{j} + 3\hat{k})$
(c) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$ (d) $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$

15. For a vector function $\vec{F}$ there exist a scalar potential only if

- (a) $\text{div } \vec{F} = 0$ (b) $\text{curl } \vec{F} = 0$
(c) $\text{grad}(\text{div } \vec{F}) = \vec{0}$ (d) $\vec{F} \cdot (\text{curl } \vec{F}) = 0$

16. A vector field which has a vanishing divergence is called as

- (a) Solenoidal field (b) irrotational field (c) rotational field (d) scalar field

17. $\vec{F} = (x + 2y + 4z)\hat{i} + (2ax - 3y - z)\hat{j} + (4x - y + 2z)\hat{k}$ is

- (a) Solenoidal (b) irrotational
(c) rotational (d) both solenoidal and irrotational

18. A unit tangent vector to the surface $x = t, y = e^t, z = -3t^2$ at $t = 0$ is

- (a) $\hat{i} + \hat{j}$ (b) $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ (c) $\frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j}$ (d) none of these

19. Unit normal vector to the surface $z = 2xy$ at the point $(2, 1, 4)$ is

- (a) $\frac{1}{\sqrt{20}}(2\hat{i} + 4\hat{j})$ (b) $\frac{1}{\sqrt{20}}(2\hat{i} - 4\hat{j})$
(c) $\frac{1}{\sqrt{21}}(2\hat{i} + 4\hat{j} - \hat{k})$ (d) $\frac{1}{\sqrt{21}}(2\hat{i} + 4\hat{j} + \hat{k})$

20. Maximum value of the directional derivative of $\phi = xyz^2$ at the point $(1,0,3)$ is

- (a) 9 (b) 10 (c) 0 (d) none of these

21. Work done by a particle along the straight line $y = x$ from $A(0,0)$ to $B(1,1)$ under the force $\vec{F} = x^2\hat{i} + xy\hat{j}$ is _____

- (a) $\frac{1}{3}$ (b) 1 (c) $\frac{4}{3}$ (d) $\frac{2}{3}$

22. If $\vec{F} = (4xy - 3x^2z^2)\hat{i} + 2x^2\hat{j} - 2x^3z\hat{k}$ then $\int_C \vec{F} \cdot d\vec{r}$ is

- (a) Independent of path (b) not defined (c) not equal to zero (d) none of these

23. The value of $\int \text{grad}(x + y - z) \cdot d\vec{r}$ from $(0,1,-1)$ to $(1,2,0)$ is

- (a) 0 (b) 3 (c) -1 (d) not obtainable

24. A necessary and sufficient condition that the line integral $\int_C \vec{F} \cdot d\vec{r}$ for every closed C vanishes is

- (a) $\text{div}\vec{F} = 0$ (b) $\text{curl}\vec{F} = \vec{0}$ (c) $\text{div}\vec{F} \neq 0$ (d) $\text{curl}\vec{F} \neq \vec{0}$

25. If $\phi(x, y, z) = c$ is a surface, then $\nabla \phi$ is

- (a) parallel to the surface (b) normal to the surface
(c) unit vector (d) none of these

26. In cylindrical coordinate system, $r = a$, where a is a constant represents

- (a) a plane (b) a cylinder with radius a
(c) a cone (d) none of these

27. In spherical coordinate system, $\theta = a$, where a is a constant represents

- (a) a plane (b) a cylinder with radius a
(c) a sphere (d) a cone

28. Spherical coordinates (r, θ, ϕ) and cartesian coordinates (x, y, z) are related as

$x =$ _____, $y =$ _____, $z =$ _____

- (a) $r\sin\phi\cos\theta, r\sin\phi\sin\theta, r\cos\phi$ (b) $r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta$
(c) $r\cos\theta\cos\phi, r\cos\theta\sin\phi, r\sin\theta$ (d) none of these

29. Cylindrical coordinates (r, θ, z) and cartesian coordinates (x, y, z) are related as

$x =$ _____, $y =$ _____, $z =$ _____

- (a) $r\sin\theta, r\cos\theta, z$ (b) not defined
 (c) $r\cos\theta, r\sin\theta, z$ (d) none of these

30. $\int \vec{F} \cdot d\vec{r}$ is independent of the path of integration if

- (a) $\vec{F}$ is conservative (b) $\vec{F} = \nabla \phi$, ϕ is a scalar function
 (c) $\oint_C \vec{F} \cdot d\vec{r} = 0$ (d) all the (a), (b), (c) are true

31. By the Green's theorem, value of $\oint_C Mdx + Ndy$ is

- (a) $\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy$ (b) $\iint_R \left(\frac{\partial N}{\partial y} - \frac{\partial M}{\partial x} \right) dxdy$
 (c) $\iint_R \left(\frac{\partial M}{\partial x} - \frac{\partial N}{\partial y} \right) dxdy$ (d) $\iint_R \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dxdy$

32. Using the Green's theorem the value of the line integral $\int_C (y^2 dx + x^2 dy)$, where C is the boundary of the square $-1 \leq x \leq 1, -1 \leq y \leq 1$ is

- (a) 0 (b) $2(x + y)$ (c) 4 (d) $\frac{4}{3}$

33. Value of $\int xdy - ydx$ around the circle $x^2 + y^2 = 1$ is

- (a) 0 (b) $\frac{1}{2}$ (c) 1 (d) 2π

34. By the Stoke's theorem, relation between the line and the surface integral is

- (a) $\int_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \cdot \vec{F}) \cdot d\vec{s}$ (b) $\int_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{s}$
 (c) $\int_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{s}$ (d) $\int_C \vec{F} \cdot d\vec{r} = \iint_S \vec{F} \cdot d\vec{s}$

35. By the Gauss divergence theorem, relation between the surface and the volume integral is

- (a) $\iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dV$ (b) $\iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \times \vec{F} dV$
 (c) $\iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \times \vec{F} dV$ (d) none of these

36. If $\vec{F} = ax\hat{i} + by\hat{j} + cz\hat{k}$, then $\int_S \vec{F} \cdot d\vec{s}$, S being the surface of a unit sphere, is

- (a) $\frac{4\pi}{3} (a + b + c)$ (b) $\frac{4\pi}{3} (a + b + c)^2$
 (c) $4\pi(a + b + c)$ (d) none of these

37. The Gauss divergence theorem uses which of the following operations?

- (a) Gradient (b) Curl (c) Divergence (d) Laplacian

38. Evaluate the Surface integral $\iint (3x\hat{i} + 2y\hat{j}) \cdot d\vec{s}$, where S is the sphere given by $x^2 + y^2 + z^2 = 9$

- (a) 60π (b) 120π (c) 180π (d) 240π

39. The Gauss divergence theorem converts

- (a) Line to surface integral (b) Line to volume integral
(c) surface to line integral (d) Surface to volume integral

40. The Stoke's theorem uses which of the following operations ?

- (a) Gradient (b) Curl (c) Divergence (d) Laplacian

41. Which of the following theorem converts line integral to surface integral ?

- (a) Gauss divergence and Stoke's theorem (b) Stoke's theorem only
(c) Green's theorem only (d) Stoke's and Green's theorem

42. Find the value of Stoke's theorem for $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$. the state of the function will be

- (a) Solenoidal (b) divergent (c) rotational (d) irrotational

43. The Stoke's theorem can be used to find which of the following ?

- (a) Area enclosed by a function in the given region (b) Linear distance
(c) Volume enclosed by a function in the given region (d) Curl of the function

ANSWERS:

- | | | |
|-------|-------|-------|
| 1. b | 22. a | 43. a |
| 2. c | 23. b | |
| 3. d | 24. b | |
| 4. c | 25. b | |
| 5. c | 26. b | |
| 6. b | 27. d | |
| 7. b | 28. b | |
| 8. a | 29. c | |
| 9. d | 30. d | |
| 10. b | 31. a | |
| 11. a | 32. a | |
| 12. b | 33. d | |
| 13. a | 34. c | |
| 14. a | 35. b | |
| 15. b | 36. a | |
| 16. a | 37. c | |
| 17. d | 38. c | |
| 18. b | 39. d | |
| 19. c | 40. b | |
| 20. a | 41. d | |

21. d

42. d

Complex Analysis

1	If $z = a$ is a pole of order m of $f(z)$ then the residue of $f(z)$ at $z = a$ is : a. $\frac{1}{(m-1)!} \lim_{z \rightarrow a} \frac{d^{m-1}}{dz^{m-1}} \{(z-a)^m f(z)\}$ b. $\frac{1}{m!} \lim_{z \rightarrow a} \frac{d^m}{dz^m} \{(z-a)^m f(z)\}$ c. $\frac{1}{(m-1)!} \lim_{z \rightarrow a} \frac{d}{dz} \{(z-a)^m f(z)\}$ d. $\frac{1}{(m-1)!} \lim_{z \rightarrow a} \frac{d^{m-1}}{dz^{m-1}} f(z)$
2	If $f(z) = \frac{z+1}{(z^2+1)^2(4z^2-1)}$ then the poles of $f(z)$ are : a. ± 1 of order 2 and simple poles at $\pm \frac{1}{2}$ b. Only simple poles at $\pm i$ and $\pm \frac{1}{2}$ c. Simple poles at $\pm i$ and $\pm \frac{1}{2}i$ d. Poles of order 2 at $\pm i$ and simple poles at $\pm \frac{1}{2}$
3	If $f(z) = u + iv$ where $z = x + iy$ then $\int_C f(z) dz$ can be expressed as : a. $\int_C (u dx + v dy) - i \int_C (v dx + u dy)$ b. $\int_C (u dx - v dy) + i \int_C (u dy + v dx)$ c. $\int_C (u dx - v dy) + i \int_C (v dx + u dy)$ d. $\int_C (u dx + v dy) + i \int_C (v dx - u dy)$
4	If $f(z) = \frac{z^2}{z-1}$ then $z = 1$ is known as : a. Zero of $f(z)$ b. Pole of $f(z)$ c. Singularity of $f(z)$ d. Residue of $f(z)$
5	A point $z = a$ where $f(z)$ fails to be analytic is known as : a. Zero of $f(z)$ b. Pole of $f(z)$ c. Singularity of $f(z)$ d. Residue of $f(z)$
6	If $f(z) = u + iv$ is an analytic function, then a. u and v are harmonic conjugates of each other b. Orthogonal to each other c. Both a. and b. d. Neither of a. nor b.
7	If $f(z) = \frac{e^z}{(z^2+1)}$ then the poles of $f(z)$ are : a. ± 1 b. $\pm i$ c. Double pole at $z = -1$ d. Simple pole at $z = -1$
8	If $f(z) = e^z$ where $z = x + iy$ then $f(z) = u + iv$ in terms of x and y is : a. $e^x \cos y + i e^x \sin y$ b. $e^x \sin y + i e^x \cos y$ c. $e^x \cos y - i e^x \sin y$ d. $e^y \cos x + i e^y \sin x$
9	A circle centered at a point $z = a$ and having radius r is denoted by : a. $z = r$ b. $z + a = r$ c. $z - a = r$ d. $z - a = r$
10	$w = \frac{az+b}{cz+d}$ is a bilinear transformation if : a. $ad - bc = 0$ b. $ab - cd \neq 0$ c. $ad - bc \neq 0$ d. $ac - bd \neq 0$
11	If C is a circle $z = 2$ then the value of $\int_C \frac{dz}{z^2-1}$ is

	a. πi b. $2\pi i$ c. 1 d. 0
12	If the function $f(z)$ satisfy Cauchy-Riemann equations, then the derivative of $f(z)$ is : a. $u_x - iv_x$ b. $v_x + iu_x$ c. $u_x + iv_x$ d. $v_x - iu_x$
13	If $f(z)$ is analytic and $f'(z)$ is continuous at all points in the region bounded by the simple closed curves C_1 and C_2 then : a. $\oint_{C_1} f(z)dz \neq \oint_{C_2} f(z)dz$ b. $\oint_{C_1} f(z)dz = \oint_{C_2} f(z)dz$ c. $\oint_{C_1} f(z)dz + \oint_{C_2} f(z)dz = 0$ d. $\oint_{C_1} f'(z)dz = \oint_{C_2} f'(z)dz$
14	A function $u(x, y)$ is said to be harmonic if : a. $u_{xx} + u_{yy} = 0$ b. $u_{xy} + u_{yx} = 0$ c. $u_{xx} - u_{yy} = 0$ d. $u_{xx} = u_{yy}$
15	u and v are harmonic conjugates of each other. If $u = x^2 - x - y^2$ then the function v is : a. $2xy - y + x + c$ b. $2x - 1 + c$ c. $x^2 + y^2 + c$ d. $2xy - y + c$
16	The poles of $f(z) = \frac{e^z}{z^2 + a^2}$ are : a. $\pm 2i$ b. 0, 1 c. $\pm ai$ d. $\pm a$
17	The poles of $f(z) = \frac{1}{(z^2 + 1)^3}$ are : a. $\pm 3i$ b. $\pm i$ c. 1, 1 d. ± 1
18	For the function $f(z) = \frac{z^2 + 3}{z^2(z + 4)}$ the pole $z = 0$ has order : a. 1 b. 0 c. 4 d. 2
19	The simple poles of $f(z) = \frac{z^2 - 4}{z^2 + 5z + 4}$ are : a. 1, 4 b. 1, -4 c. -1, -4 d. -1, 4
20	A function $\Phi(x, y)$ satisfying the Laplace equation is known as : a. Analytic b. Holomorphic c. Harmonic d. Non – Harmonic
21	A function $f(z)$ satisfying the Cauchy-Riemann equation is known as : a. Analytic b. Non – Analytic c. Harmonic d. Non – Harmonic
22	The function e^z is : a. Analytic everywhere b. Analytic no where c. Only differentiable d. Not differentiable
23	$w = \log z$ is analytic everywhere except : a. On the real axis b. at $z = \pm 1$ c. At $\pm i$ d. At $z = 0$
24	If $f(z) = u + iv$ is an analytic function then :

d. $2\pi i \sum \text{residues at each poles}$

Answers:

1 - a	2 - d	3 - c	4 - b	5 - c
6 - c	7 - b	8 - a	9 - d	10 - c
11 - d	12 - c	13 - b	14 - a	15 - d
16 - c	17 - b	18 - d	19 - c	20 - c
21 - a	22 - a	23 - d	24 - c	25 - c
26 - d	27 - d	28 - a	29 - b	30 - b
31 - b	32 - a	33 - c	34 - c	35 - d

Fourier Analysis Z-transform

- Which of the following is not Dirichlet's condition for the Fourier series expansion?
 - $f(x)$ is periodic, single valued, finite
 - $f(x)$ has finite number of discontinuities in only one period
 - $f(x)$ has finite number of maxima and minima
 - $f(x)$ single valued, finite
- What is the Fourier series expansion of the function $f(x)$ in the interval $(c, c + 2\pi)$?
 - $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$
 - $\sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$
 - $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$
 - $1 + \sum_{n=0}^{\infty} a_n \cos(nx) + \sum_{n=0}^{\infty} b_n \sin(nx)$
- If the function $f(x)$ is even, then which of the following is zero in the Fourier series expansion of $f(x)$?
 - a_n
 - b_n
 - a_0
 - nothing is zero
- If the function $f(x)$ is odd, then which of the only coefficient is present in the Fourier series expansion of $f(x)$?
 - b_n
 - a_n
 - a_0
 - nothing is zero
- If $Z^{-1}\left(\frac{z}{z-a}\right)$ is a^n , then $Z^{-1}\left(\frac{2z}{2z+a}\right)$ is :
 - $(-2a)^n$
 - $\left(-\frac{2}{a}\right)^n$
 - $\left(-\frac{a}{2}\right)^n$
 - $-2a^n$
- If $Z(1) = \frac{z}{z-1}$ then $Z(2)$ is :
 - $\frac{z}{z-2}$
 - $\frac{2z}{z-1}$
 - $\frac{2z}{z-2}$
 - $\frac{z}{2(z-1)}$
- The value of b_n in the Fourier series expansion of the function $f(x) = x^2$ in $(-\pi, \pi)$.
 - Non zero finite value
 - infinite series
 - can't be found
 - zero
- The Fourier series expansion of an odd periodic function contains only ____
 - Sine terms
 - Constant term
 - Cosine terms
 - Both constant term and cosine terms
- In half range cosine Fourier series, we assume the function to ____
 - Odd function
 - Even function
 - Can't be determined
 - Can be any thing
- The Fourier series of an even function of time does not have ____
 - Cosine terms
 - Constant term
 - Sine terms
 - Constant term and cosine terms

11. If $Z(u_n) = U(z)$ then $\lim_{z \rightarrow \infty} U(z)$ is :

- a. 0 b. ∞ c. u_0 d. u_∞

12. $Z^{-1}\left(\frac{1}{z+a}\right)$ is :

- a. $(-a)^{n-1}$ b. $-a^{n-1}$ c. $(-1)^{n-1}a$ d. $\frac{1}{a}$

13. If u_n is a function of n then the bilateral Z -transform of u_n is defined as,

$$Z(u_n) =$$

- a) $\sum_{n=0}^{\infty} u_n z^n$ b) $\sum_{n=0}^{\infty} u_n z^{-n}$
c) $\sum_{n=-\infty}^{\infty} u_n z^{-n}$ d) $\sum_{n=-\infty}^{\infty} u_n z^n$

14. If $f(x)$ is defined in $(-\pi, \pi)$, then $a_0 =$ _____

- a) $\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$ b) $\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$
c) $\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$ d) $\int_{-\pi}^{\pi} f(x) dx$

15. . Formulae for evaluation of a_2 in Fourier series expansion of the give set of point

$(x_i, y_i): i = 0, 1, 2 \dots n$ is _____

- a) $\frac{1}{n} \sum y \cos 2x$ b) $\frac{2}{n} \sum y \cos 2x$ c) $\frac{1}{n} \sum y \sin 2x$ d) $\frac{2}{n} \sum y \sin 2x$

16. Which of the following function is odd

- a) $f(x) = \begin{cases} 1+x, & -\pi < x < 0 \\ 1-x, & 0 < x < \pi \end{cases}$ b) $f(x) = \begin{cases} 1+\frac{2x}{\pi}, & -\pi < x < 0 \\ 1-\frac{2x}{\pi}, & 0 < x < \pi \end{cases}$
c) $f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$ d) $f(x) = \begin{cases} -x, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$

17. The period of $\cos 3x$ is _____

- a) 2π b) $\frac{2\pi}{3}$ c) $\frac{\pi}{6}$ d) $\frac{4\pi}{3}$

18. The value of b_n in the Fourier series of the function $f(x) = 1$ in the interval $(-1, 1)$ is

- a) 0 b) $\sum_{n=0}^{\infty} \frac{2}{n} \sin \frac{n\pi x}{2}$ c) $\sum_{n=0}^{\infty} \frac{2}{n^2} \cos \frac{n\pi x}{2}$ d) 1

19. If $f(x) = 4x^3$ in $(-1, 1)$, then the Fourier coefficient a_n is _____

- a) Non zero finite value b) infinite series c) zero d) can't be found

20. A function $f(x)$ defined for $0 < x < 1$ can be extended to an odd periodic function

- in _____
- a) Harmonic series b) Half range Fourier cosine series
c) Half range Fourier sine series d) cosine series
21. If $f(x) = x^4$ in $(-3, 3)$, then the Fourier coefficient b_n is _____
a) Non zero finite value b) infinite series c) zero d) can't be found
22. If $f(x)$ is defined in $(-\pi, \pi)$, then $a_n =$ _____
a) $\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$ b) $\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$
c) $\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$ d) $\int_{-\pi}^{\pi} f(x) \cos nx dx$
23. If $f(x) = x^2$ in $(-\pi, \pi)$, then the Fourier coefficient a_0 is _____
a) Non zero finite value b) Infinite series c) Zero d) Can't be found
24. If $f(x) = \begin{cases} -x, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$, then $f(x)$ is an _____ in $(-\pi, \pi)$.
a) Periodic function b) Odd function c) Even function d) Harmonic function
25. If $f(x)$ is in $(-c, c)$, then $b_n =$ _____
a) $\frac{1}{c} \int_{-c}^c f(x) dx$ b) $\frac{1}{c} \int_{-c}^c f(x) \cos nx dx$
c) $\frac{1}{c} \int_{-c}^c f(x) \sin nx dx$ d) $\int_{-c}^c f(x) \sin nx dx$
26. If the Fourier series expansion of the function $f(x) = \sqrt{1 - \cos x}$ in the interval $0 < x < 2\pi$ is $\frac{2\sqrt{2}}{\pi} - \frac{4\sqrt{2}}{\pi} \sum_{n=1}^{\infty} \frac{\cos nx}{(2n-1)(2n+1)}$ then the sum of the infinite series $\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots$ is
a) $\frac{\pi^2}{12}$ b) 1 c) $\pi + \pi^2$ d) $\frac{1}{2}$
27. The value b_n in the Fourier series expansion of $f(x) = |x|$ in $(-\pi, \pi)$ is _____
a) Zero b) Infinite sine series c) Infinite cosine series d) Can't be found
28. The Half range Fourier sine series expansion of $f(x) = 1$ in $(0, \pi)$ has only _____
a) Cosine terms b) Constant terms only
c) Sine terms d) constant and sine terms
29. Period of the function $f(x) = |\sin x|$ is _____
a) 2π b) π c) $\frac{\pi}{2}$ d) None of these
30. The Half range Fourier sine series expansion of the function $f(x)$ in $(0, 1)$ is _____
a) $\sum_{n=1}^{\infty} b_n \sin(nx)$ b) $\sum_{n=1}^{\infty} b_n \sin(n\pi x)$
c) $\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right)$ d) $\sum_{n=1}^{\infty} b_n \sin(2n\pi x)$

ANSWERS:

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (a) | 2. (c) | 3. (b) | 4. (a) | 5. (c) |
| 6. (b) | 7. (d) | 8. (a) | 9. (b) | 10. (c) |
| 11. (c) | 12. (a) | 13. (b) | 14. (a) | 15. (b) |
| 16. (a) | 17. (b) | 18. (a) | 19. (c) | 20. (c) |
| 21. (c) | 22. (b) | 23. (a) | 24. (c) | 25. (c) |
| 26. (d) | 27. (a) | 28. (c) | 29. (b) | 30. (d) |